

Implementation Priority Guide

Consolidated Across the OFM-JEPA / Reflow-JEPA / Monotonic-OT-HTR Document Set

Mosa

August 2026

How to use this document

Twelve files were reviewed for this pass: the CFM system-design/test-plan, two revisions of the Monotonic-OT HTR master guide, two revisions of the OFM-JEPA technical report, two revisions of the Reflow-JEPA proof document, the General OT-VL-JEPA proposal, an architecture-blueprint markdown note, and my own earlier two-coupling design note from this conversation. This guide does three things: (1) corrects a conflation error from earlier in this conversation, (2) ranks the work into three implementation priorities, and (3) for each priority, gives the exact architecture, the math required to understand *why* it is built that way, and the concrete next engineering step.

Correction. Earlier in this conversation I produced a "two-coupling" design that imported a patch $\leftrightarrow$ character many-to-many alignment problem into the VQA architecture. Having now read the actual source documents: **that problem does not exist in** . transports a single pooled image-question embedding to a single answer embedding — there is no patch-sequence-to-character-sequence step anywhere in it. The patch $\leftrightarrow$ character many-to-many correspondence is a real problem, but it belongs entirely to the *separate* Arabic HTR line of work (Monotonic OT / Sinkhorn alignment, Priority 2 below). The two projects share a family of tools (optimal transport, flow matching) but are different pipelines solving different tasks. Treat my earlier design note as superseded by Priority 1 below for , and by Priority 2 below for the HTR alignment problem.

1 Priority ranking and why

	What	Status	Why this rank
P1	Reflow-JEPA (CFM+Reflow VQA), from the system-design/test-plan doc	Design-complete; 31/31 tests passing against config-matched random-weight architectures; not yet trained	The only document in the set explicitly marked implementation-ready, with every prior failure mode traced to a corrected fix and validated in code
P2	Monotonic OT + Flow Matching for Arabic HTR	Theory and proofs complete (corrected revision); no code yet	Independent, real, and solid theory, but one engineering layer behind P1 — no test suite yet, and it feeds a different active project (FOTOPIA HTR work)
P3	General OT-VL-JEPA (ICNN / Schrödinger Bridge / Sinkhorn paradigms)	Superseded proposal; kept as escalation path	Already decided against in favor of Reflow-JEPA’s CFM+Reflow approach; relevant only if P1’s ablations later demand it

Implement in this order: P1 first (it is genuinely ready — the remaining work is swapping in real checkpoints and writing the training loop already specified in §2.4 of its source document). P2 second, on its own timeline, since it is a different task. P3 is not an implementation target now; it is documented here only so that if P1’s ablations (already specified in its own test plan) show the flow field misbehaving in a way Reflow can’t fix, you know exactly which of the three rejected paradigms to reach for and why.

2 Priority 1 — Reflow-JEPA: Conditional Flow Matching for VQA

2.1 Task and mechanism

VQA is treated as conditional domain transfer: an ODE moves a point through a shared latent space from a (visual, question) representation to an answer representation. No autoregressive token generation; decoding is nearest-neighbor snap onto a pre-encoded candidate answer bank. $d_{\text{shared}} = 768$ throughout, once past the projection heads.

2.2 Component table (exact, from the design doc)

Component		Choice	Status	Output
Visual encoder E_V		I-JEPA ViT-H/14, 224×224	Frozen	$H_v \in \mathbb{R}^{[B, 256, 1280]}$
Question encoder E_C		T5-base / BART-base encoder	Frozen	$c \in \mathbb{R}^{[B, 768]}$
Q-Pool		$K=8$ learned query slots, FiLM(c)-modulated, cross-attend over H_v	Trainable	$z_v^{\text{raw}} \in \mathbb{R}^{[B, 1280]}$
Visual projection g_V		2-layer MLP + ViCReg	Trainable	$\tilde{z}_v \in \mathbb{R}^{[B, 768]}$
Text answer encoder E_T (online)		T5-base / BART-base encoder	Trainable	$z_t^{\text{raw}} \in \mathbb{R}^{[B, 768]}$
Text answer encoder E'_T (target)		EMA of E_T , same modality	Stop-grad; used for $\mathcal{M}_T$ only	—
Text projection g_T/g'_T		2-layer MLP + ViCReg (online/target pair)	Trainable / EMA	$\tilde{z}_t \in \mathbb{R}^{[B, 768]}$
Stochastic source		$Z_0 = \tilde{z}_v + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$	New mechanism	$Z_0 \in \mathbb{R}^{[B, 768]}$
Predictor v_θ		6-layer DiT, sinusoidal+AdaLN time, cross-attn on $\tilde{z}_v$ and c	Trainable	$v_{\text{pred}} \in \mathbb{R}^{[B, 768]}$
Candidate bank $\mathcal{M}_T$		$\{g'_T(E'_T(a_k))\}$, pre-encoded via target (EMA) copy	Cached	vectors in $\mathbb{R}^{768}$
Decoder		Cosine nearest-neighbor snap onto $\mathcal{M}_T$	Rule-based	answer string

2.3 Training loss and procedure

$$Z_\tau = (1 - \tau)Z_0 + \tau Z_1, \quad \tau \sim \text{Unif}[0, 1], \quad \mathcal{L}_{\text{total}} = \|v_{\text{pred}} - (Z_1 - Z_0)\|^2 + \lambda(\mathcal{L}_{\text{var}}(\tilde{z}_v) + \mathcal{L}_{\text{var}}(\tilde{z}_t))$$

Phase 1 (base CFM). Train v_θ, g_V, g_T (online) jointly on $\mathcal{L}_{\text{total}}$; update E'_T, g'_T as EMA of the online text pipeline every step; E_V, E_C frozen throughout.

Phase 2 (Reflow). For $k = 1, \dots, K$: simulate $(Z_0^{(k)}, \hat{Z}_1^{(k)})$ pairs by integrating the current field; retrain $v_{\theta_{k+1}}$ on the resulting deterministic coupling; log the straightness/crossing diagnostic $\hat{S} + \hat{V}$ each round; stop once it plateaus.

Inference. Cache $\tilde{z}_v^*, c^*$ per (image, question). Draw a *fresh* ε every call; integrate $dz/d\tau = v_\theta(z_\tau, \tau, \tilde{z}_v^*, c^*)$ from $\tau=0$ to $\tau=1-\delta$ ($\delta \sim 10^{-3}$); snap to the nearest candidate in $\mathcal{M}_T$ via cosine

similarity. For multi-answer-coverage evaluation, draw multiple independent ε per (image, question) — one draw returns one atom, not the answer distribution.

2.4 The math you need, in the order the design depends on it

1. Why not plain L^2 regression (Lemma 1).

Lemma 1 (Off-manifold collapse). *The minimizer of $\mathcal{L}_{JEPA}(\theta) = \mathbb{E}\|f_\theta(z_v, c) - z_t\|^2$ is $f_\theta^*(z_v, c) = \mathbb{E}[z_t \mid z_v, c]$. If $\mathcal{M}_T$ is non-convex (true for a finite answer vocabulary) and $P_T(\cdot \mid z_v, c) = (1 - \lambda)\delta_a + \lambda\delta_b$ for distinct answers a, b , then $f_\theta^*(z_v, c) = (1 - \lambda)a + \lambda b \notin \mathcal{M}_T$ on a positive-probability set of prompts.*

This is why the predictor cannot be a bare regressor: on any genuinely ambiguous question it outputs a spatial centroid that is not itself a valid answer embedding.

2. Why CFM fixes it (marginal preservation).

Theorem 1 (Marginal preservation). *Under realizability (v_{θ_1} attains the exact population minimizer of $\mathcal{L}_{CFM}$) and well-posedness (Picard–Lindelöf conditions on v_{θ_1}), $\text{Law}(Z_\tau \mid c) = \text{Law}(z_\tau \mid c)$ for every τ . At $\tau=1$: $\hat{z}_t \mid c \sim P_T(\cdot \mid c)$ exactly, so $\hat{z}_t \in \mathcal{M}_T$ almost surely.*

Proof idea: both processes solve the same continuity equation $\partial_\tau p_\tau + \nabla \cdot (v_{\theta_1}(\cdot, \tau, c) p_\tau) = 0$ with the same initial condition, and the solution is unique under the stated hypotheses. This is the reason the design decodes by nearest-neighbor snap rather than regression: the flow’s terminal marginal is provably supported on the answer manifold, not just near it in expectation.

3. Why the source must be stochastic, and why that’s not hand-wavy. A deterministic source ($Z_0 \equiv \tilde{z}_v$, a point mass) forces the one-step field back onto exactly the off-manifold compromise point Lemma 1 warns about: $v_\theta(\tilde{z}_v, 0, c) = \mathbb{E}[z_t \mid \tilde{z}_v, c] - \tilde{z}_v$ reproduces the failure verbatim. The fix, $Z_0 = \tilde{z}_v + \sigma\varepsilon$, is not just plausible — it has a closed form. By Brenier’s theorem plus semi-discrete optimal transport (Aurenhammer 1987; Aurenhammer–Hoffmann–Aronov 1998), the optimal map from a continuous (Gaussian) source to a discrete target partitions the source space into *Laguerre cells*, one per target atom; each cell maps entirely onto its atom. In the two-atom case this reduces to a half-space split of ε -space, so which atom a given draw lands on is explicit and checkable. This closed form is what test_07 (below) validates numerically against 200,000 Monte Carlo draws.

Two independent things stochasticity is doing here, not one: (a) it recovers correct per-example multimodal answers (proved via the Laguerre-cell argument above), and (b) it does *not* by itself prevent representation collapse of g_V, g_T — with both constant, the source noise is exactly invertible from Z_τ , so zero CFM loss is achievable despite total collapse. This is why VICReg is a separate, independently necessary term, not a redundant safety margin.

4. Terminal-time behavior near $\tau = 1$ (corrected rate). For the explicit single-atom-cell field $v(z, \tau) = (a - z)/(1 - \tau)$:

$$\nabla_z v(z, \tau) = -\frac{I_d}{1 - \tau} \implies \|\nabla_z v(z, \tau)\|_{\text{op}} = \frac{1}{1 - \tau} \quad (\text{linear, not cubic}).$$

Integrating to an early-stopping cutoff $\tau = 1 - \delta$:

$$\int_0^{1-\delta} \|\nabla_z v(z, \tau)\|_{\text{op}} d\tau = -\ln \delta \quad (\text{logarithmic, not super-exponential}).$$

This is the corrected version of an earlier (fabricated/unverifiable) claim that the Jacobian scaled as $(1 - \tau)^{-3}$ with a Grönwall factor $\exp(1/2\delta^2)$; the real rate matches the well-documented $O(1/t)$ score-singularity behavior in diffusion models near the data manifold. Practical consequence: a small, non-vanishing stopping margin ($\delta \sim 10^{-3}$) is sufficient — no exotic architecture is needed near $\tau = 1$.

5. Reflow: what it provably does and does not do.

Theorem 2 (Non-increasing convex transport cost). *For any convex γ with $\mathbb{E}[\gamma(z_t - z_v)] < \infty$, $\mathbb{E}[\gamma(\hat{z}_t - z_v)] \leq \mathbb{E}[\gamma(z_t - z_v)]$.*

Theorem 3 (Best-iterate straightening, $O(1/K)$). $\min_{0 \leq k < K} [S(Z^{(k+1)}) + V(Z_0^{(k)}, Z_1^{(k)})] \leq \mathbb{E}\|z_t - z_v\|^2/K$.

This is a best-iterate guarantee, not monotone convergence: it guarantees *some* round with small straightness+crossing exists within K rounds, not that round K itself is best, and not that crossings hit exactly zero.

Proposition 1 (Straightness is necessary but not sufficient for exact OT, $d \geq 2$). *A zero-crossing fixed point of Reflow need not be the Brenier-optimal map once $d \geq 2$: a pure rotation is Lipschitz, deterministic, and non-crossing, yet is not optimal for a rotation-invariant Pr . Exact identification with the quadratic-cost OT map additionally requires the learned field to be a gradient field with a convex potential.*

Practical reading: don’t expect Reflow alone to hand you the literal OT map in high dimensions; it gives you a straighter, cheaper-to-integrate field with a provable cost bound, which is what the design actually needs (fewer NFEs at inference), not a claim of exact optimality.

2.5 Pre-implementation test suite (31 tests, currently passing)

Ordered so failures are caught closest to their source: frozen encoders first, then cross-encoder compatibility, then the custom trainable modules, then the decoder, then the core theoretical mechanism decoupled from any network at all.

Suite	What it locks down
01 visual encoder	Output shape $(B, 256, 1280)$ exact; correct CLS-token handling (true I-JEPA has none); no representation collapse; finite non-zero norms; frozen $\Rightarrow$ zero gradient
02 text encoder	Output shape $(B, 768)$; deterministic re-encoding; distinct answers stay separated (cosine sim < 0.999); no collapse; reports (doesn’t gate on) intrinsic dimension of $\mathcal{M}_T$
03 encoder compatibility	Confirms projection heads are mandatory (raw dims don’t already match); measures the real $\ z_i\ $ vs. $\ z_t\ $ norm ratio; confirms L2-normalization equalizes it
04 Q-Pool	Output shape $(B, 768)$; gradients reach every one of the K slots; output changes with image; output changes with question (confirms FiLM is load-bearing, not decorative)
05 VICReg collapse guard	Collapsed batch $\Rightarrow$ penalty $\approx \gamma_0^2$; healthy batch $\Rightarrow$ penalty ≈ 0 ; strict ordering between them; partial (dimension-wise) collapse is detected, not just total collapse
06 decoder hubness	Raw Euclidean NN shows severe hubness at realistic bank size; prescribed cosine decoding roughly halves the skew; NICDM cuts Euclidean skew by $> 50\%$; NICDM-on-cosine is measured but not assumed to help (documented negative/inconclusive result)
07 stochastic-source theorem	<i>The single most important test:</i> 200,000-draw Monte Carlo mixture matches the target λ within 6-sigma binomial tolerance; zero samples land near the compromise point; deterministic-source baseline reproduces the original failure explicitly for contrast
08 terminal divergence	Measured Jacobian norm matches the linear $1/(1 - \tau)$ rate via autograd; integrated error matches $-\ln \delta$ to $< 1\%$; $\delta = 10^{-3}$ keeps integrated rate < 10

2.6 Concrete next steps

1. Swap `confest.py`'s two factory functions to `.from_pretrained(...)` with the real I-JEPA ViT-H/14 and T5/BART checkpoints.
2. Re-run test 01–03 and read the printed diagnostics (norm ratio, intrinsic dimension) — these real numbers, not the mock-weight numbers, should set σ , confirm or revise $d_{\text{shared}}=768$, and decide whether hubness needs a correction beyond cosine.
3. Only then assemble the full training loop from §2.4 (Phase 1 base CFM, then Phase 2 Reflow).

2.7 Open items (unresolved by design, need real data/checkpoints)

- σ (source noise scale): unset; tune against manifold-adherence and multi-answer-coverage metrics.
- $d_{\text{int}}(\mathcal{M}_T)$: must be re-measured on the real checkpoint and full answer vocabulary — do not import the ≈ 7 –15 figure from Tulchinskii et al., which measures sentence-level word embeddings for AI-text detection, not VQA answer embeddings.
- NICDM-on-cosine: unresolved on synthetic data; re-measure on the real trained candidate bank.
- K (Reflow rounds) and the early-stopping margin δ : need a real sweep per the base theory's convergence protocol (track validation transport cost, the $\hat{S}+\hat{V}$ diagnostic, and downstream VQA accuracy together; accuracy should track the diagnostic curve, not keep climbing on unrelated compute).

3 Priority 2 — Monotonic Optimal Transport for Arabic HTR

3.1 Task and mechanism (a different pipeline from Reflow-JEPA)

HTR maps M visual feature frames (columns extracted along a handwritten line image) to N character tokens, $M \neq N$ in general. This is the project where a patch $\leftrightarrow$ character many-to-many correspondence genuinely exists.

3.2 Core math, in dependency order

1. Why Monge fails discretely.

Definition 1 (Push-forward). $(T_{\#}\mu)(B) := \mu(T^{-1}(B))$.

Monge seeks $\inf_T \int_{\mathcal{X}} c(x, T(x)) d\mu(x)$ s.t. $T_{\#}\mu = \nu$. For discrete $\mu = \sum_i a_i \delta_{x_i}$, $\nu = \sum_j b_j \delta_{y_j}$ with uniform mass ($a_i=1/M$, $b_j=1/N$), a non-splitting map exists iff $N \mid M$. When $M=100$ frames and $N=30$ characters (say), $100/30 \notin \mathbb{Z}$ and mass splitting is mandatory — Monge has zero feasible solutions. This is the formal version of the many-to-many/ligature problem.

2. Kantorovich’s relaxation and why quadratic cost. Kantorovich replaces the map with a coupling $\pi \in \Pi(\mu, \nu)$, allowing mass splitting. Under quadratic cost, Brenier’s theorem (1991) says this loses no structure *in the continuous, non-atomic case*: $\pi^* = (\text{id} \times T)_{\#}\mu$ for $T = \nabla\varphi$, convex φ .

Remark 1. *Brenier’s theorem requires μ absolutely continuous, which the M - and N -atom Dirac measures here do not satisfy. It is invoked only as the continuous-limit justification for why quadratic cost specifically enforces monotone order (in 1D, $T = \varphi'$ is monotone non-decreasing); the discrete monotonicity result below is proved independently, via an elementary exchange argument, not via Brenier.*

3. The discrete monotonicity guarantee (full proof, worth having on hand). Composite cost: $\tilde{C}_{ij} = c_{\text{sem}}(v_i, t_j) + \lambda(x_i - y_j)^2$, $x_i=i/M$, $y_j=j/N$, $c_{\text{sem}} \in [0, 2]$ (cosine distance).

Theorem 4 (Quantitative monotonicity threshold). *If $\lambda > 2MN$, the unregularized optimal coupling Π^* contains zero crossing assignments (no $i_1 < i_2$, $j_1 > j_2$ both carrying positive mass).*

Proof sketch (5 steps, full algebra in the source document). (1) For a crossing pair, $\Delta C_{\text{pos}} = 2(x_{i_1} - x_{i_2})(y_{j_1} - y_{j_2}) < 0$, via direct expansion and cancellation of squared terms. (2) Bounded above by $-2/(MN)$ at adjacent indices. (3) $\Delta C_{\text{sem}} \leq 4$ in the worst case since $c_{\text{sem}} \in [0, 2]$. (4) Requiring $\lambda \Delta C_{\text{pos}} + \Delta C_{\text{sem}} < 0$ in the worst case gives $\lambda > 2MN$. (5) Mass-redistribution contradiction: if Π^* had a crossing pair, moving mass δ to uncross it strictly lowers cost, contradicting optimality. $\square$

Remark 2. *This is a worst-case sufficient bound. Well-trained embeddings rarely span the full $c_{\text{sem}} \in [0, 2]$ range, so smaller λ typically suffices in practice — but $\lambda > 2MN$ can make the Gibbs kernel $K = \exp(-\tilde{C}/\epsilon)$ ill-conditioned for M, N in the hundreds unless ϵ is scaled jointly.*

4. Entropic regularization and the Sinkhorn solver (needed for GPU training).

$$\mathcal{L}_{\text{M-Sinkhorn}} = \min_{\Pi \in U(a, b)} \sum_{i, j} \Pi_{ij} \tilde{C}_{ij} - \epsilon H(\Pi), \quad H(\Pi) = - \sum_{i, j} \Pi_{ij} (\ln \Pi_{ij} - 1).$$

Lagrangian stationarity gives the scaling form $\Pi^* = \text{diag}(u) K \text{diag}(v)$, $K_{ij} = \exp(-\tilde{C}_{ij}/\epsilon)$, with the Sinkhorn–Knopp updates

$$u^{(k+1)} = \frac{a}{K v^{(k)}}, \quad v^{(k+1)} = \frac{b}{K^{\top} u^{(k+1)}}.$$

As $\epsilon \rightarrow 0^+$, $\Pi_{\epsilon}^* \rightarrow \arg \max_{\Pi \in \arg \min \langle \Pi, \tilde{C} \rangle} H(\Pi)$ (Cominetti & San Martín, 1994) — the max-entropy selection among unregularized optimizers. Since $\lambda > 2MN$ makes that optimizer generically a unique vertex, the entropy-selection step is generically vacuous and $\Pi_{\epsilon}^* \rightarrow \Pi^*$ directly.

5. Why not ICNNs, and why flow matching alone isn’t enough either. Input Convex Neural Networks ($T_\omega = \nabla_x \varphi_\omega$) have four structural problems for this task: a costly double-backward (Hessian-vector product) to train, non-negative weight constraints that limit capacity, minimax instability in dual formulations, and — decisively — $T_\omega : \mathbb{R}^d \rightarrow \mathbb{R}^d$ acts pointwise and *cannot change sequence length*, so it cannot perform the $M \rightarrow N$ reduction HTR requires by itself. Plain OT-CFM has the mirror problem: the ODE state z_τ must keep identical tensor shape for all τ , so continuous flow matching cannot operate directly between $V \in \mathbb{R}^{M \times d}$ and $T \in \mathbb{R}^{N \times d}$ without a prior resampling step. Neither tool alone closes the loop.

3.3 The Unified Two-Stage Hybrid Architecture (the actual target design)

Stage 1 (discrete monotonic OT): length reduction $M \rightarrow N$ and monotonic order, via the Sinkhorn coupling above.

$$S = \frac{VW_Q(E_{\text{pos}}W_K)^\top}{\sqrt{d}} \in \mathbb{R}^{M \times N}, \quad \hat{\Pi} \in U(a, b) \text{ (Sinkhorn-normalized)}, \quad Z_0 = \hat{\Pi}^\top V \in \mathbb{R}^{N \times d}.$$

Stage 2 (continuous flow refinement): warp Z_0 toward target text latents $T \in \mathbb{R}^{N \times d}$ along $Z_t = (1 - t)Z_0 + tT$ via velocity network $v_\theta(Z_t, t)$, trained by the OT-CFM regression loss.

Unified loss:

$$\mathcal{L}_{\text{Unified}} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \alpha(t) \cdot \mathcal{L}_{\text{M-Sinkhorn}}(V, T) + \beta \cdot \mathcal{L}_{\text{OT-FM}}(\theta),$$

where $\mathcal{L}_{\text{M-Sinkhorn}}$ is evaluated on the *original* (V, T) pair (it trains Stage 1’s coupling, hence Z_0), and $\mathcal{L}_{\text{OT-FM}}$ is evaluated on (Z_0, T) rows (it trains Stage 2 once Z_0 is already length-aligned). $\alpha(t)$ is a warm-up scheduler balancing the two gradient scales.

For fast inference, an **Amortized Neural Bridge** B_ω can learn the transport plan directly (bypassing $K \approx 20\text{--}50$ test-time Sinkhorn loops), in either of two parameterizations: a transport-coupling predictor (outputs N -length sequences directly, native to the $M \neq N$ mismatch) or a continuous 1D monotonic convolution acting as a direct Monge map (preserves length M , needs a CTC-style decoder downstream). The design doc is honest that its latency numbers here are projections, not benchmarks — treat them as such until measured.

3.4 Concrete next steps

1. Implement Stage 1 alone first: the composite cost matrix + Sinkhorn layer with $\lambda > 2MN$, on the cleaned KHATT/RASAM/MUHARAF data. Validate the zero-crossing property empirically before adding Stage 2.
2. Benchmark Stage 1’s alignment against the existing CTC baseline on the same data split.
3. Only then add Stage 2 (OT-CFM refinement) and the unified loss, sweeping $\alpha(t)$ and checking Gibbs-kernel conditioning under the required λ .

4 Priority 3 — Alternatives Already Considered (background, not an implementation target)

The General OT-VL-JEPA proposal considered three paradigms before CFM+Reflow (P1) was settled on. Keep this section only as an escalation reference:

- **Paradigm A — Static Brenier map via ICNN.** Rejected for the same structural reasons as in §3.5 above (double-backward cost, weight constraints, minimax instability); additionally a static map can't express the time-conditioned refinement Reflow provides.
- **Paradigm B — Diffusion Schrödinger Bridge SDE.** Not the default for Reflow-JEPA's outer transport — full embeddings are effectively continuous, so the plain OT-CFM assumption already used in P1 is plausible without paying for a stochastic bridge. **This is the correct escalation path only if** P1's own ablations (already specified in its test plan: NFE-to-quality, vector-field curvature vs. the straight-line interpolant) show the marginal field still has curvature/averaging artifacts after Reflow has been applied.
- **Paradigm C — Semi-discrete Sinkhorn-JEPA.** Superseded specifically by the stochastic-source-plus-Brenier/Laguerre-cell result in §2.4.3 above, which gives an explicit closed form for the atomic-target case without needing a separate entropic training loss.